BMS BOARD

REVISION A

STATUS: IN-DEVELOPMENT

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02 INTERFACING

REVISION HISTORY

A PRELIMINARY RELEASE

APPLICABLE DOCUMENTS

NOTES

01 THIS CIRCUIT IS FLIGHT HARDWARE

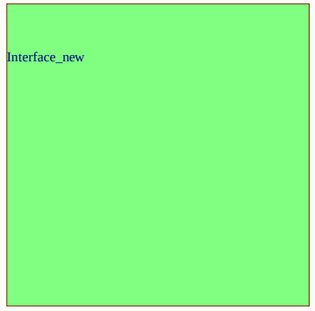
02 THIS CIRCUIT SHALL OPERATE IN TEMPERATURES FROM -20C TO 80C

03 ALL COMPONENTS DESIGNATED AS "DNP" SHALL NOT BE POPULATED DURING ASSEMBLY UNLESS OTHERWISE SPECIFIED IN ASSEMBLY DOCUMENTATION

04 ALL RESISTORS SHALL HAVE THE FOLLOWING RATINGS UNLESS OTHERWISE SPECIFIED:
> TOLERANCE OF ±1% OR LOWER
> TEMPERATURE STABILITY OF 100PPM/°C OR LOWER

05 ALL CAPACITORS SHALL HAVE THE FOLLOWING RATINGS UNLESS OTHERWISE SPECIFIED:
> TOLERANCE OF ±10% OR LOWER
> MAXIMUM VOLTAGE OF 25V OR HIGHER

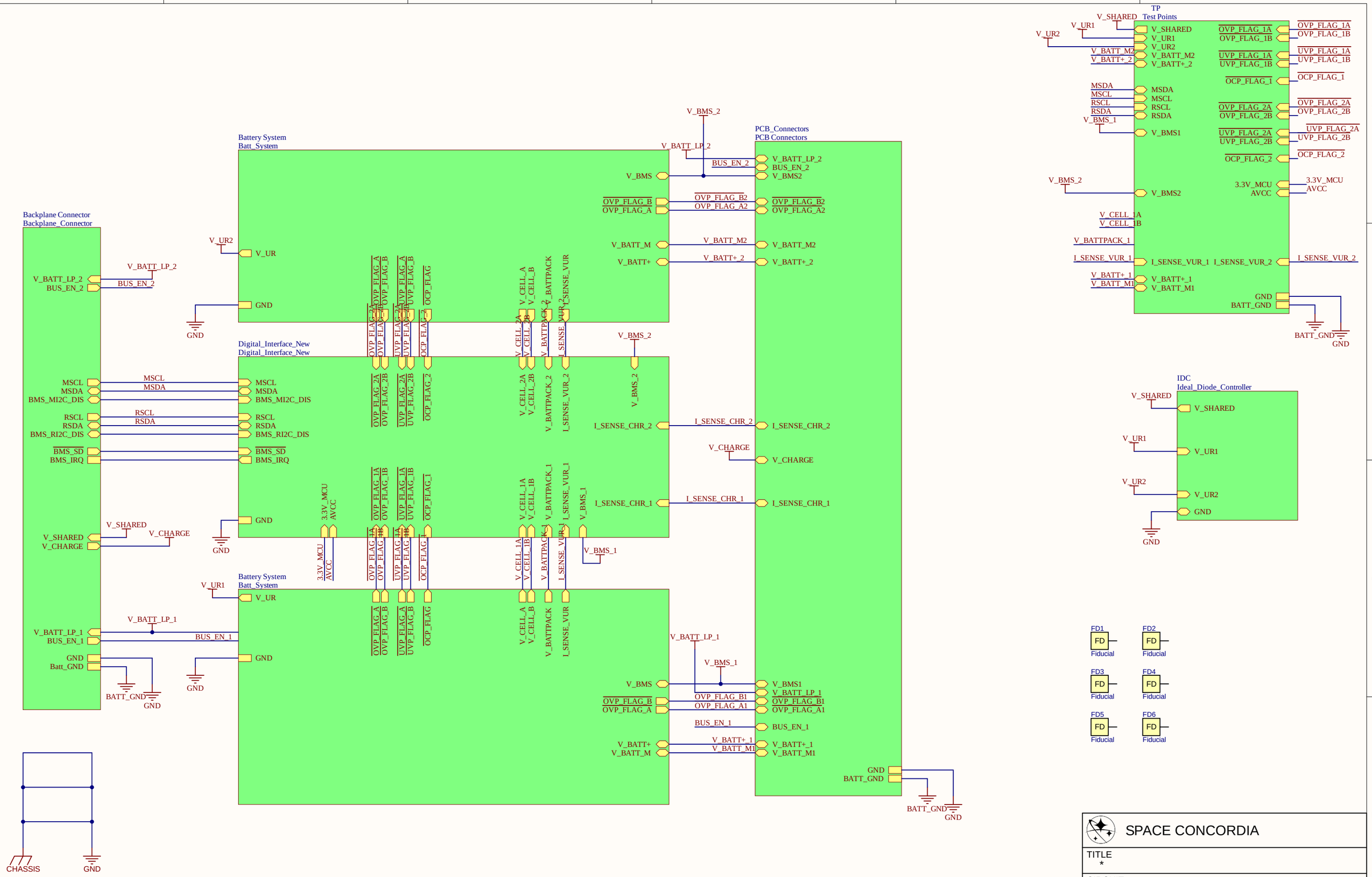
06 ALL INDUCTORS SHALL HAVE THE FOLLOWING RATINGS UNLESS OTHERWISE SPECIFIED:
> TOLERANCE OF ±20% OR LOWER



X

SPACE CONCORDIA

TITLE: COVER PAGE			
CIRCUIT: BMS BOARD			
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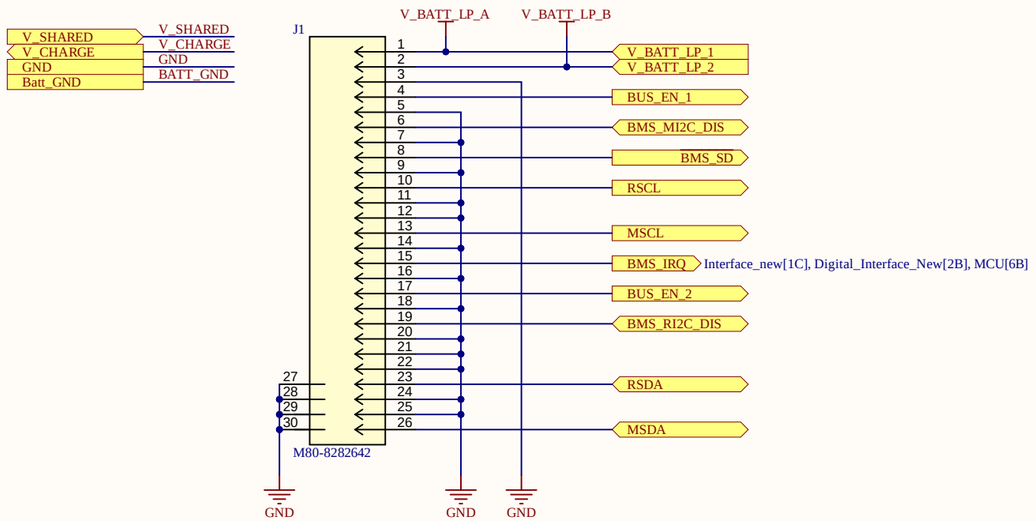


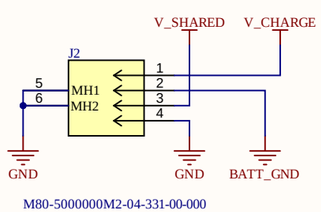
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CIRCUIT
BMS BOARD

SIZE B | REV A | SHEET 2 / 19



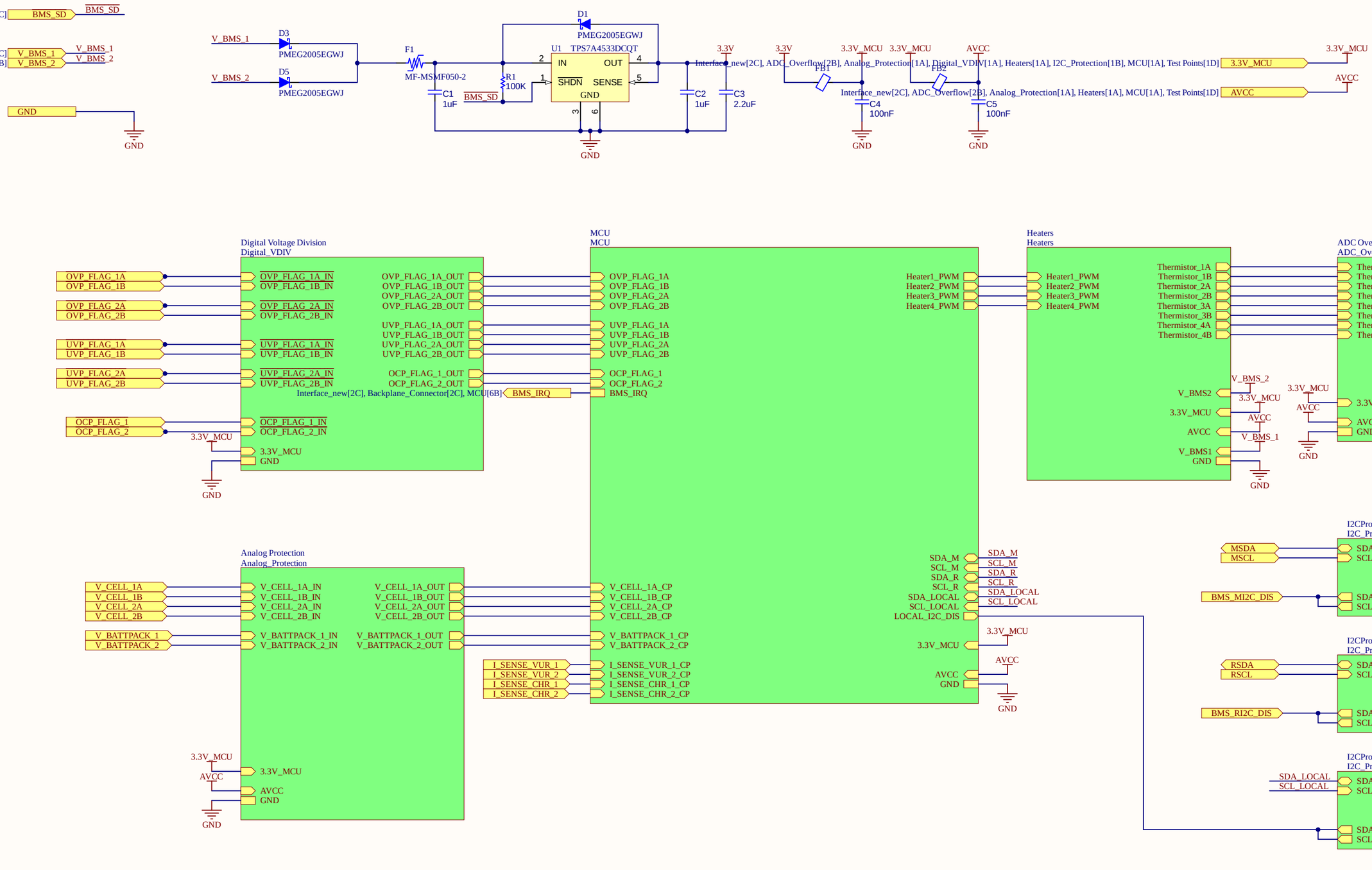


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CIRCUIT
BMS BOARD

SIZE B | REV A | SHEET 3 / 19

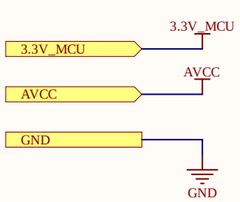


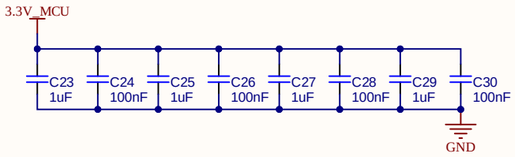
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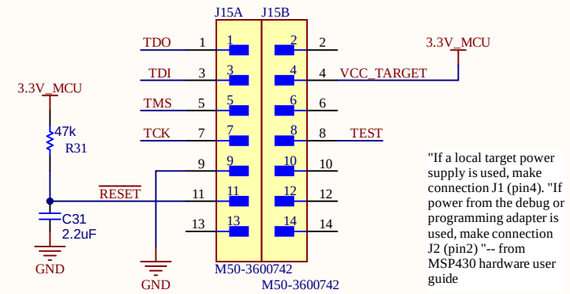
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CIRCUIT
BMS BOARD

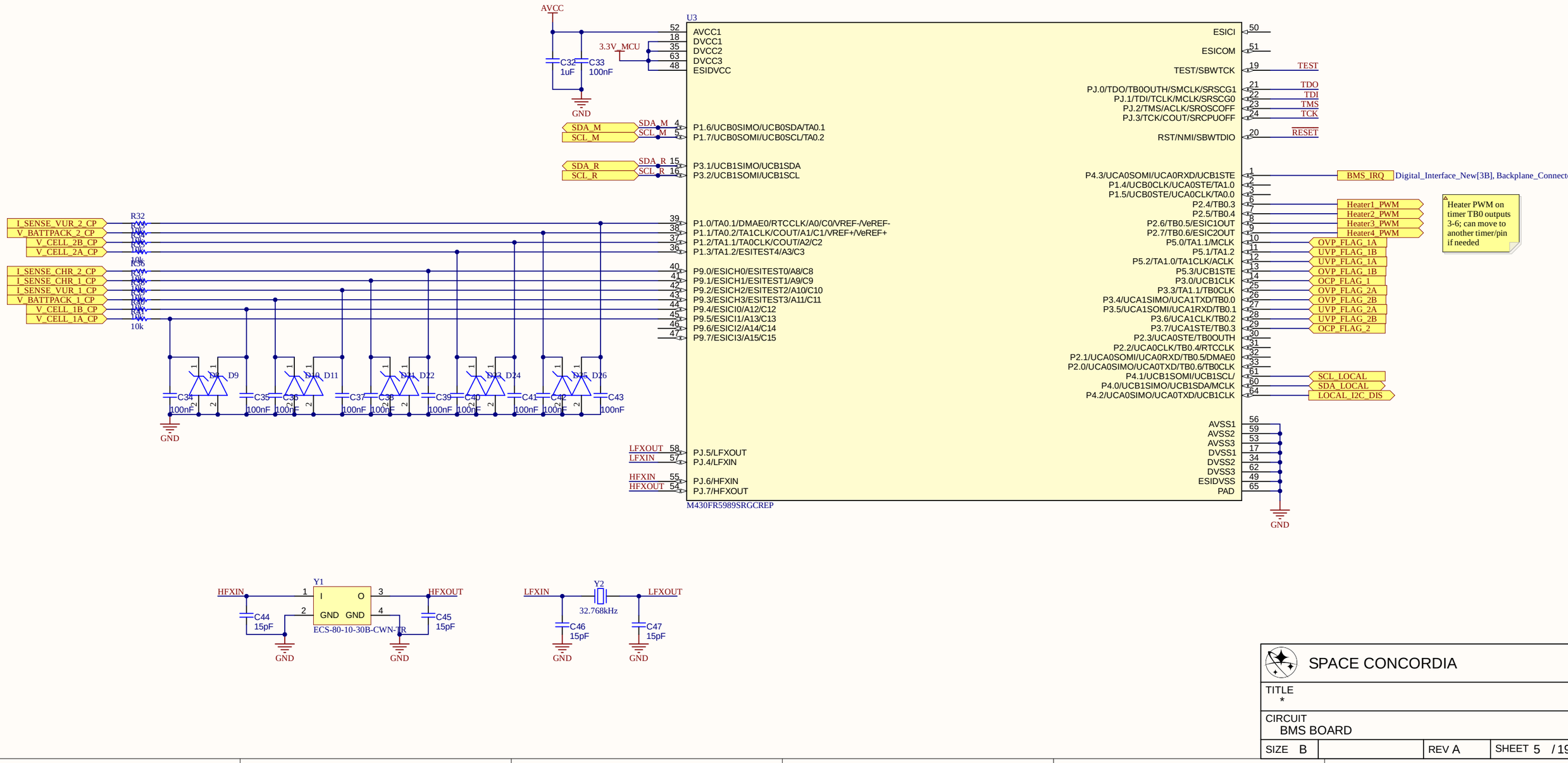
SIZE B | REV A | SHEET 4 / 19







"If a local target power supply is used, make connection J1 (pin4). "If power from the debug or programming adapter is used, make connection J2 (pin2) "-- from MSP430 hardware user guide

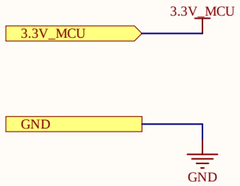


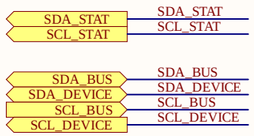
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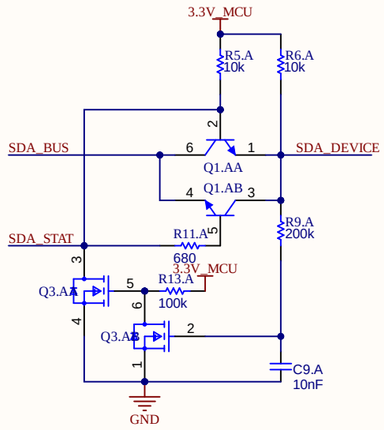
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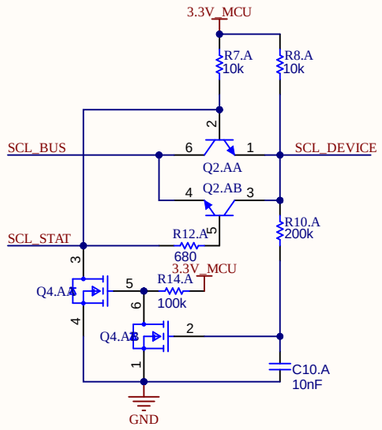
CIRCUIT
BMS BOARD

SIZE B | REV A | SHEET 5 / 19

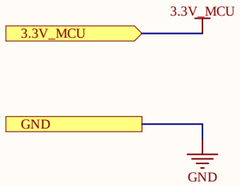


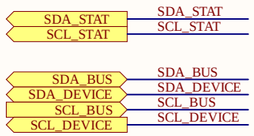


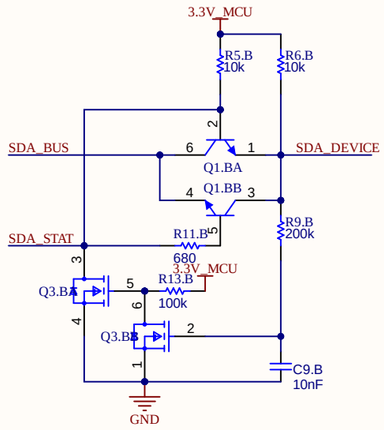


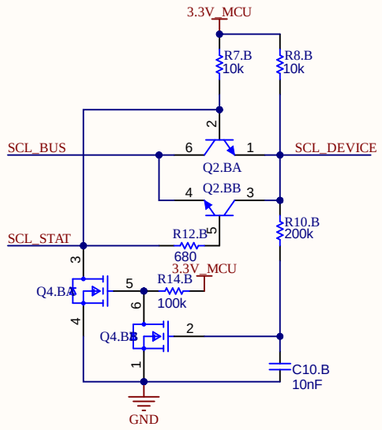


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CIRCUIT BMS BOARD			
SIZE B		REV A	SHEET 6 / 19







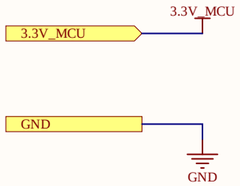


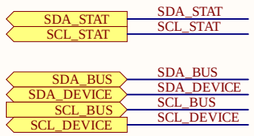
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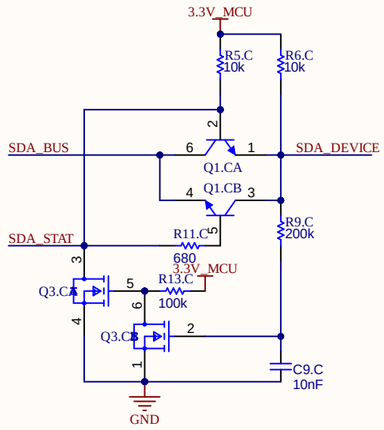
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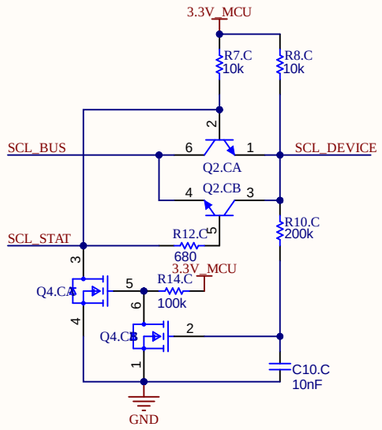
CIRCUIT
BMS BOARD

SIZE		REV	SHEET
B		A	6 / 19







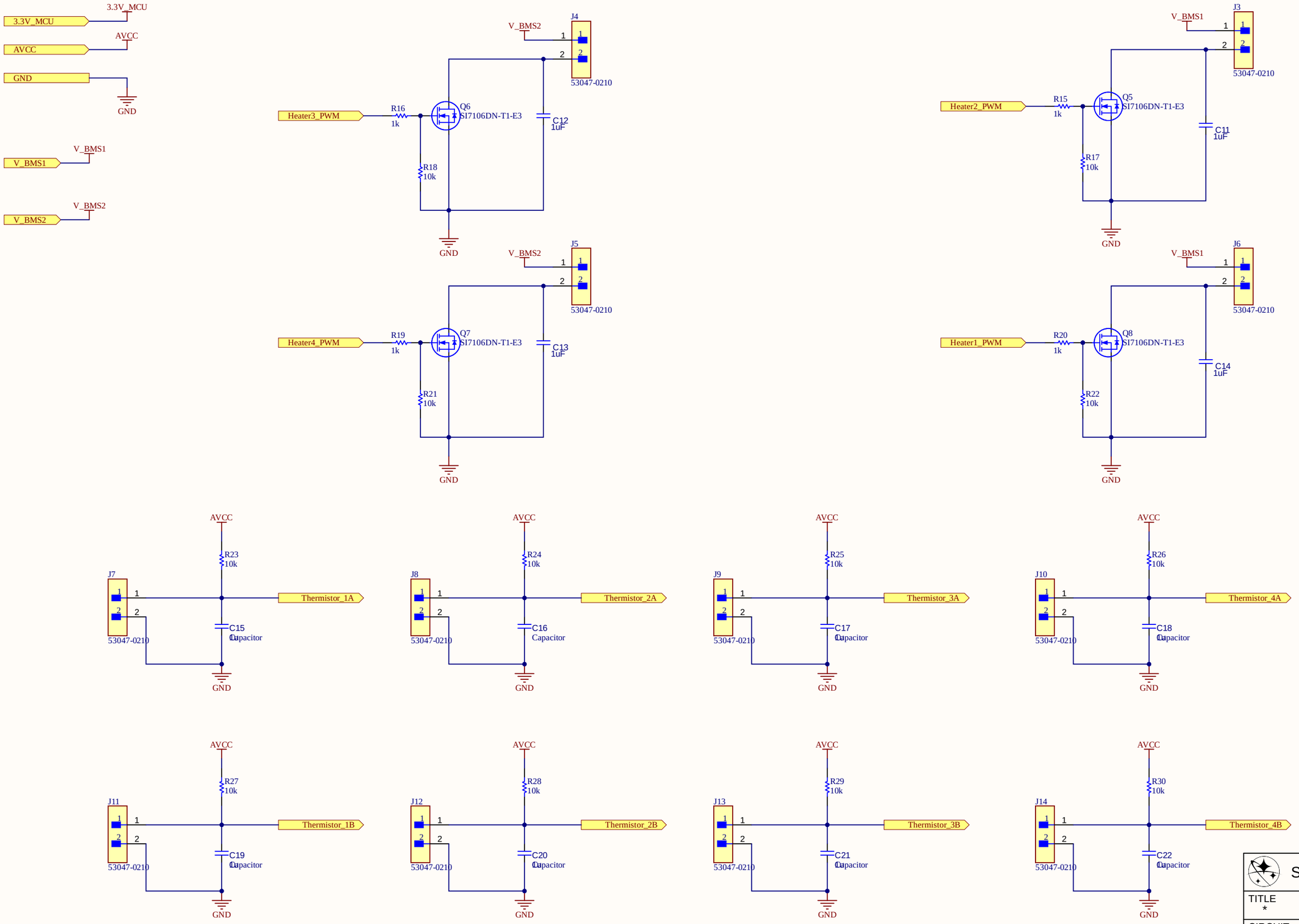


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CIRCUIT
BMS BOARD

SIZE B | REV A | SHEET 6 / 19



SPACE CONCORDIA

TITLE
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CIRCUIT
BMS BOARD

SIZE B | REV A | SHEET 7 / 19

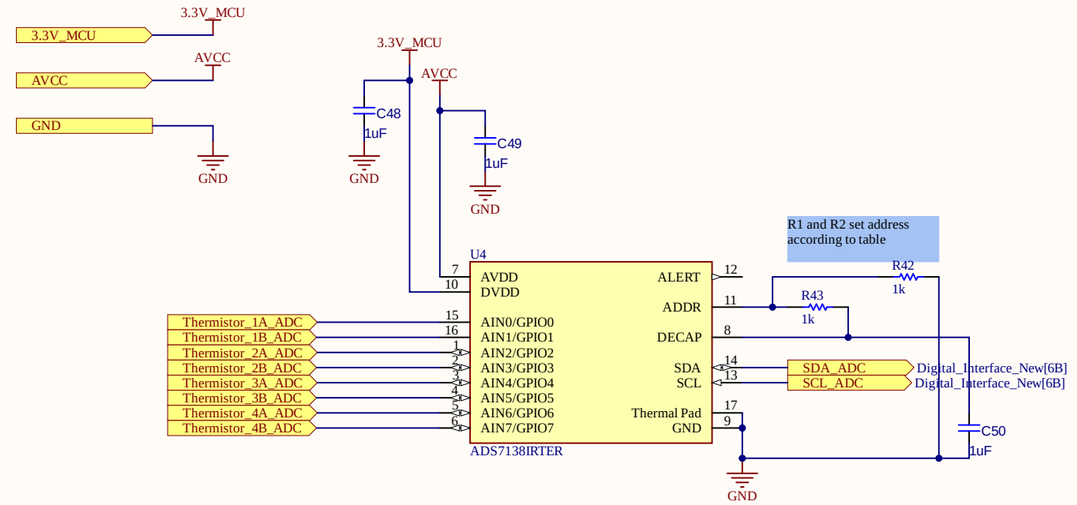


Table 2. I²C Address Selection

RESISTORS		ADDRESS
R1(1)	R2(1)	
0 Ω	DNP(2)	001 0111b (17h)
11 kΩ	DNP(2)	001 0110b (16h)
33 kΩ	DNP(2)	001 0101b (15h)
100 kΩ	DNP(2)	001 0100b (14h)
DNP(2)	DNP(2)	001 0000b (10h)
DNP(2)	11 kΩ	001 0001b (11h)
DNP(2)	33 kΩ	001 0010b (12h)
DNP(2)	100 kΩ	001 0011b (13h)

(1) Tolerance for R1, R2 ≤ ±5%.
(2) DNP = Do not populate.

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CIRCUIT
BMS BOARD

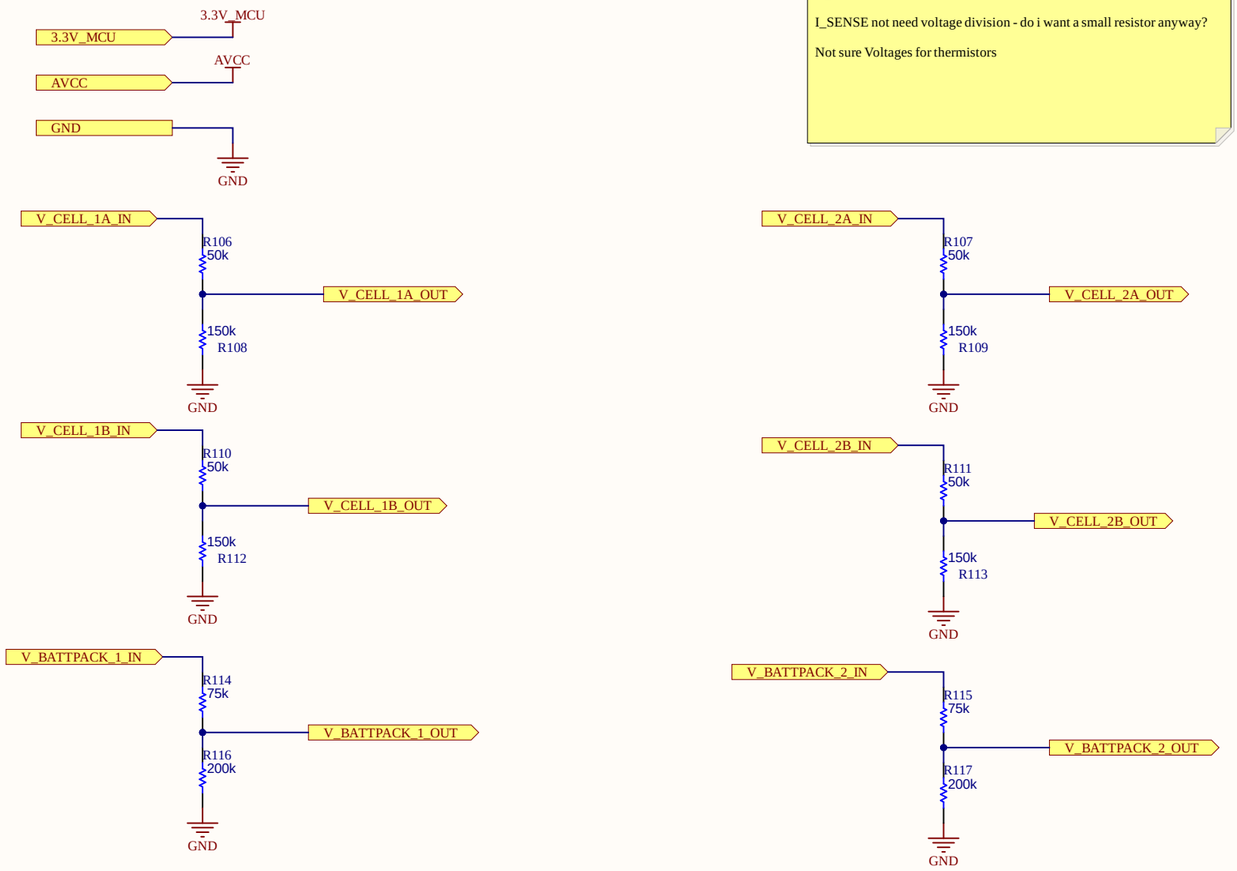
SIZE B | REV A | SHEET 8 / 19

V_CELL = 4.2 - div to 75% so 3.15

V_BATTPACK = 8.4 - div to 75% so 3.15

I_SENSE not need voltage division - do i want a small resistor anyway?

Not sure Voltages for thermistors

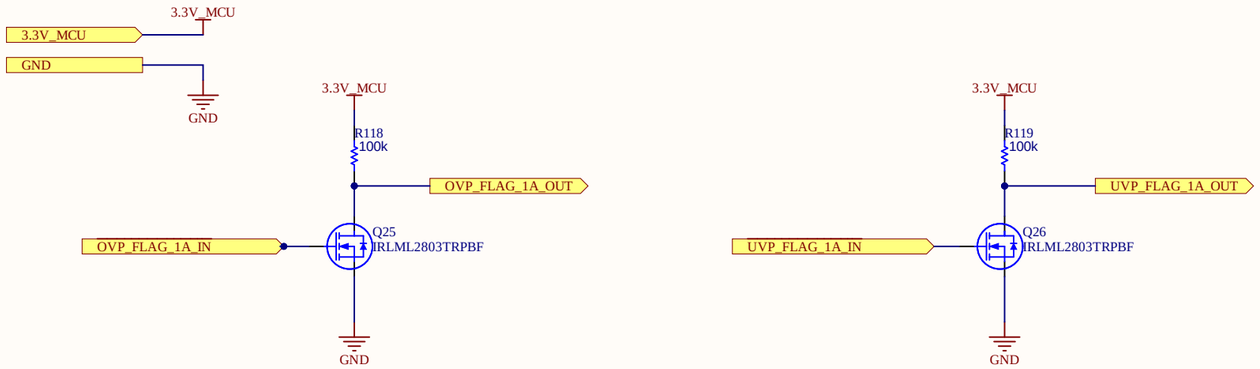


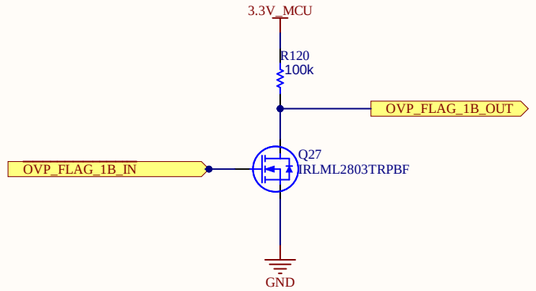
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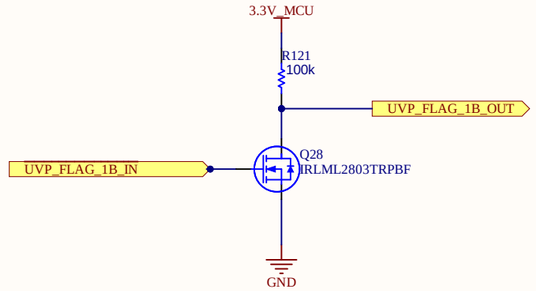
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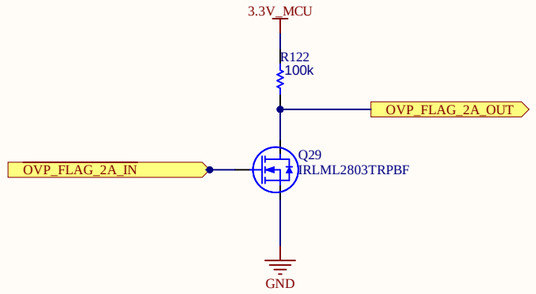
CIRCUIT
BMS BOARD

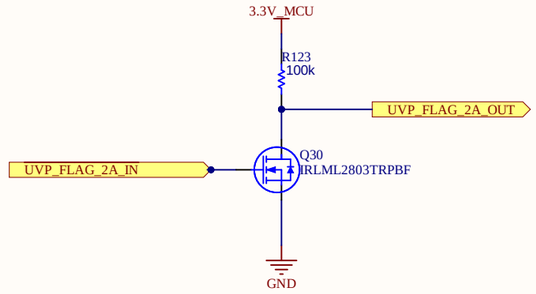
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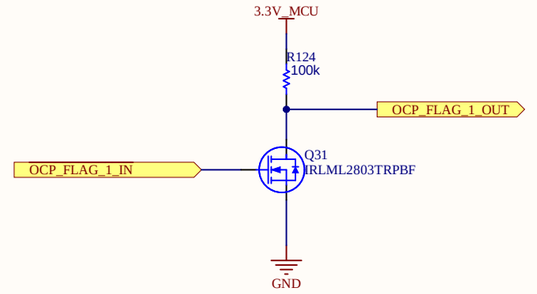


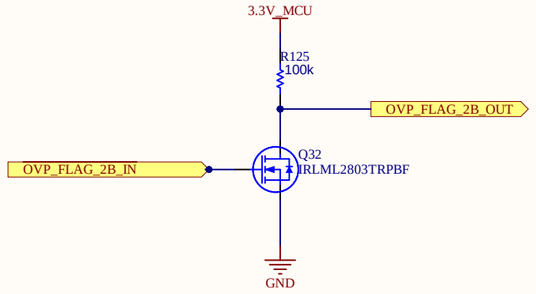


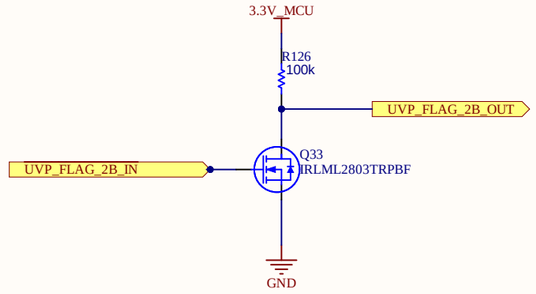


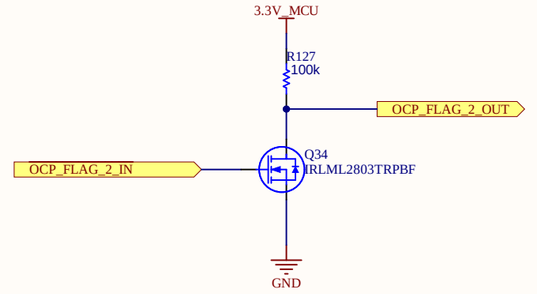












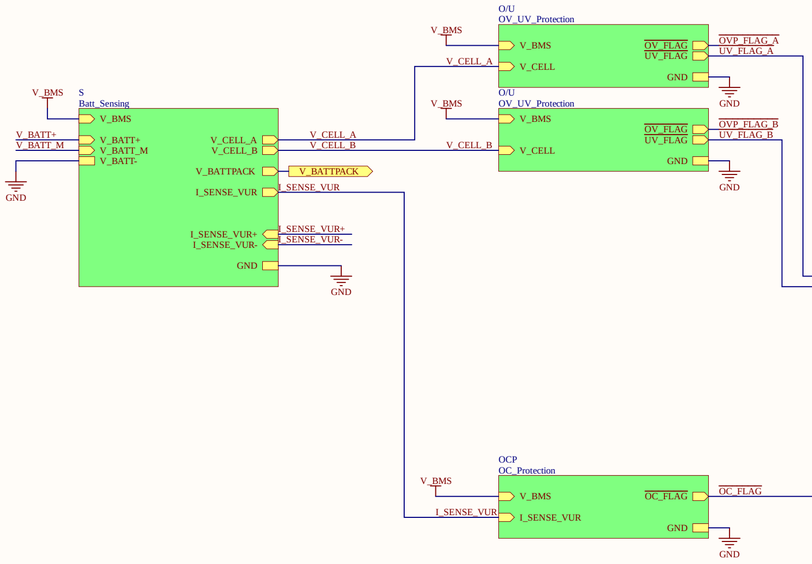
SPACE CONCORDIA

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CIRCUIT
BMS BOARD

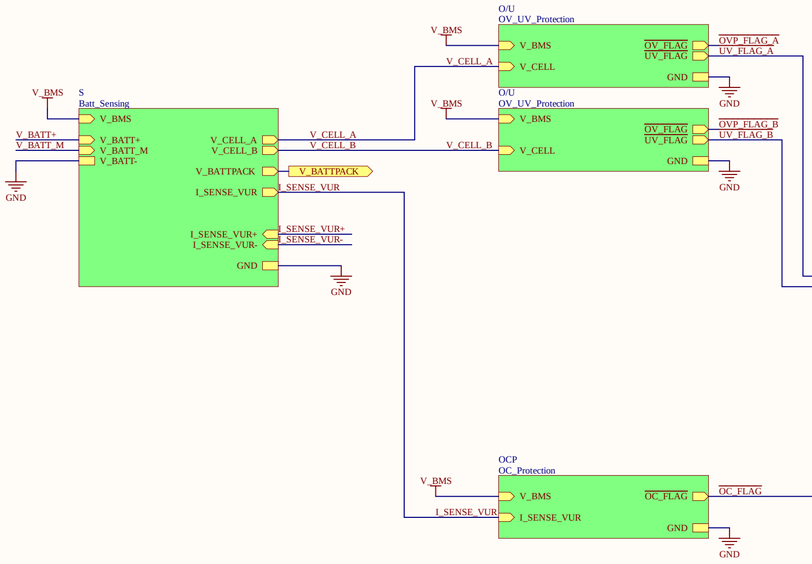
SIZE B | REV A | SHEET 10 / 19





SPACE CONCORDIA	
TITLE	*
CIRCUIT	BMS BOARD
SIZE C	REV A
SHEET	11 / 20



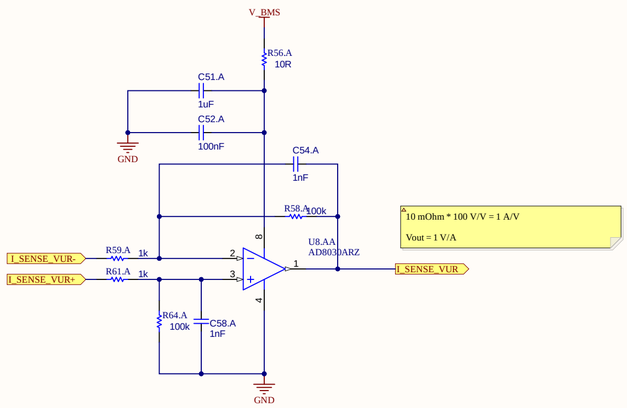


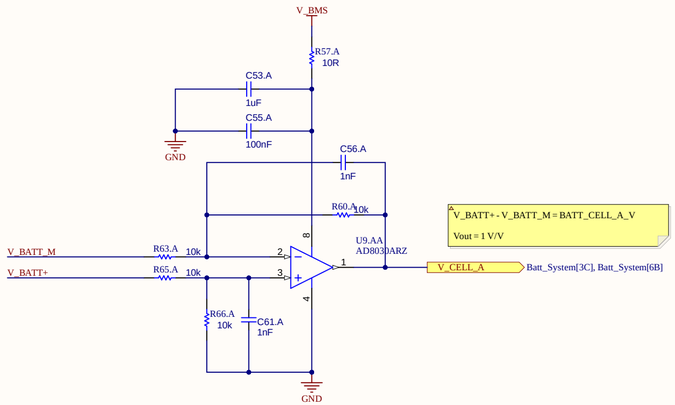
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TITLE	*
CIRCUIT	BMS BOARD
SIZE C	REV A
SHEET	11 / 20

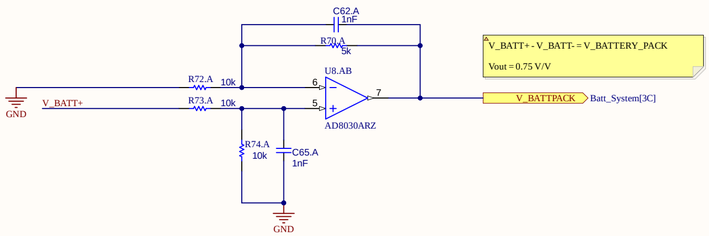
V_BATT+ V_BATT+
V_BATT_M V_BATT_M
V_BATT- V_BATT-

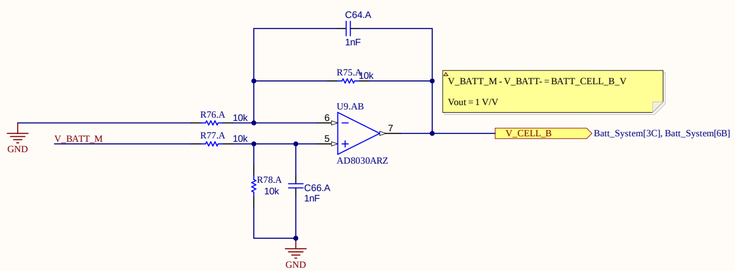
GND
GND

V_BMS
V_BMS







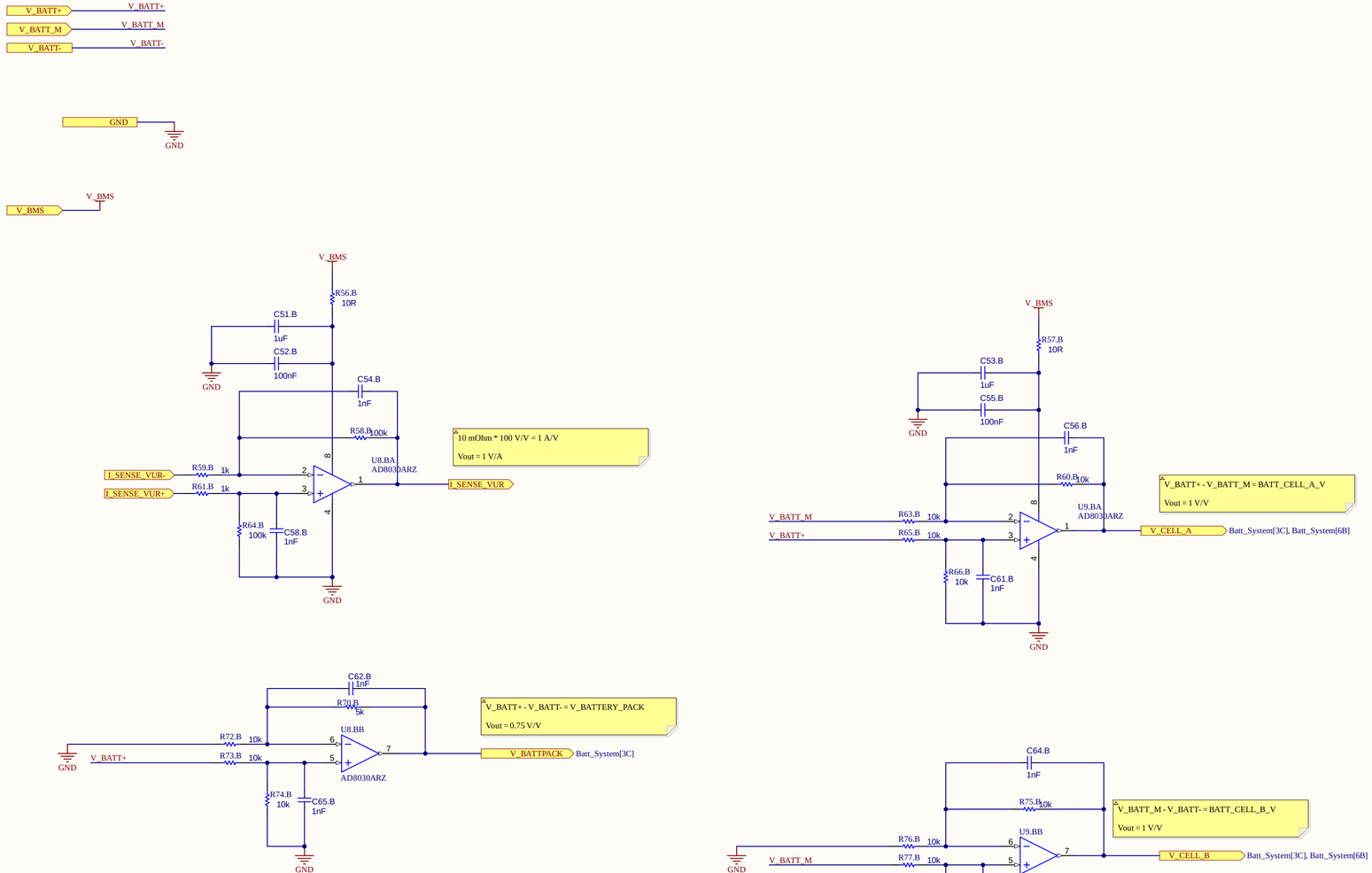


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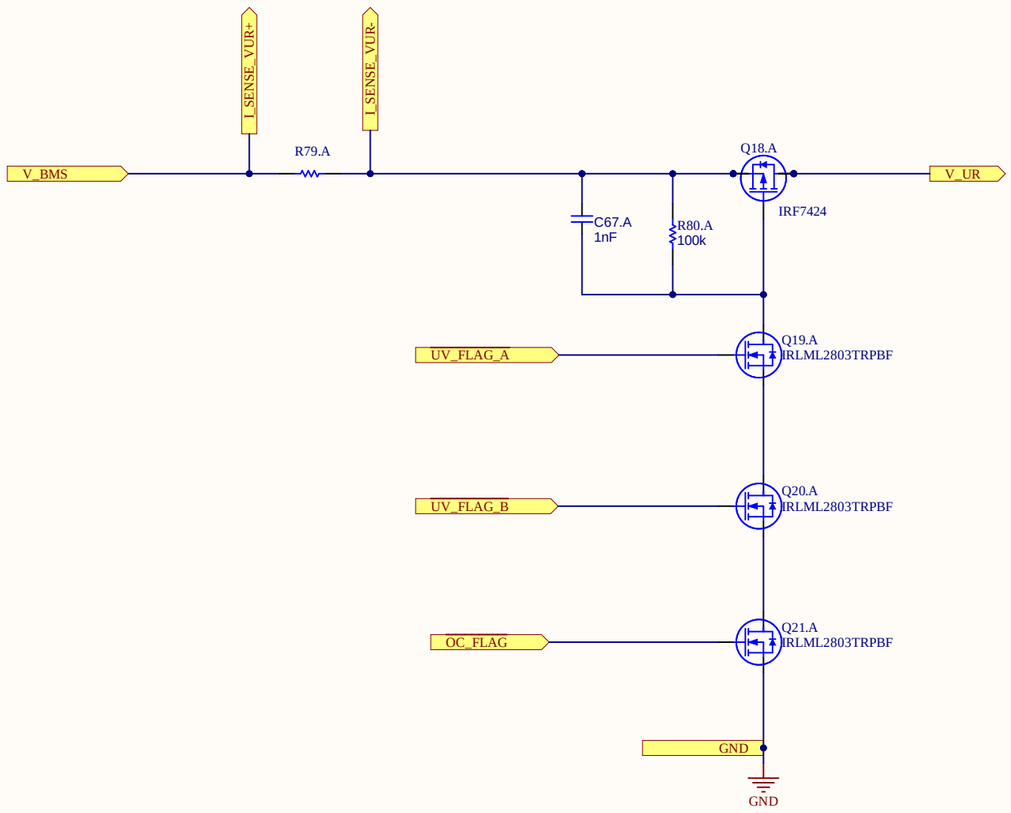
TITLE
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CIRCUIT
BMS BOARD

SIZE C | REV A | SHEET 12 / 19



SPACE CONCORDIA			
TITLE *			
CIRCUIT BMS BOARD			
SIZE C		REV A	SHEET 12 / 19



SPACE CONCORDIA

TITLE
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CIRCUIT
BMS BOARD

SIZE B | REV A | SHEET 14 / 20

Make sure can charge with RBF pin in

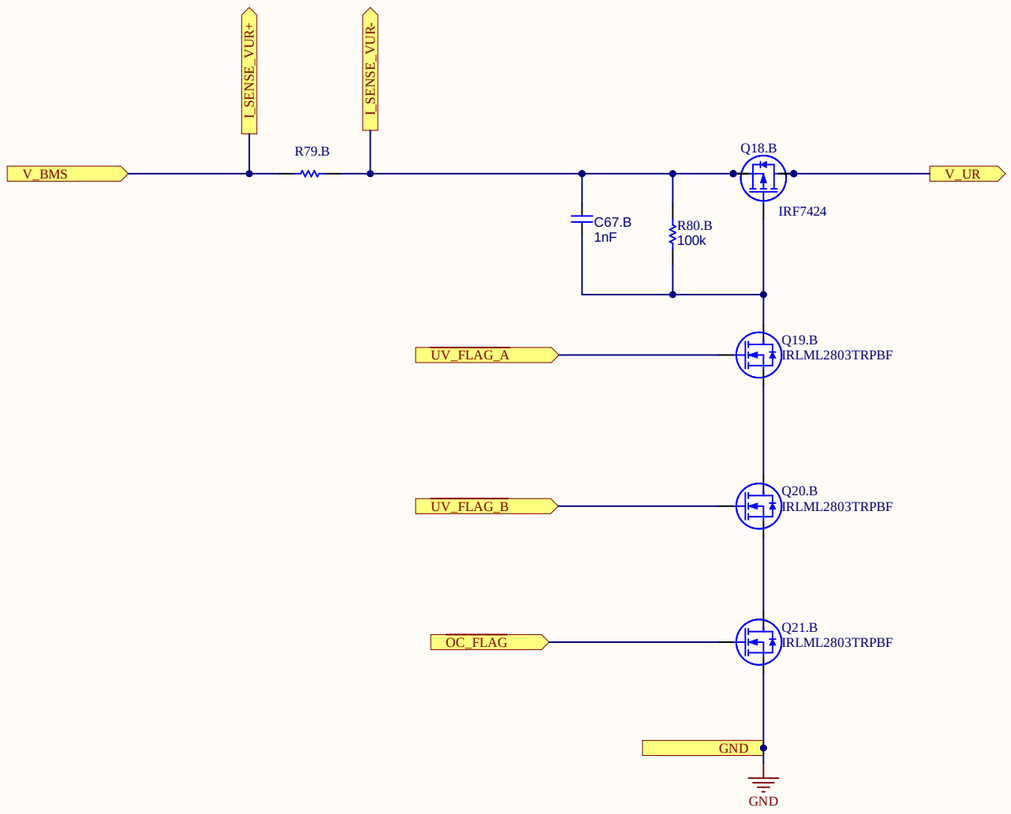
Will use 2 ideal diode controllers

SPACE CONCORDIA

TITLE
*

CIRCUIT
BMS BOARD

SIZE B | REV A | SHEET 14 / 19



SPACE CONCORDIA

TITLE
*

CIRCUIT
BMS BOARD

SIZE		REV	SHEET
B		A	14 / 20

Make sure can charge with RBF pin in

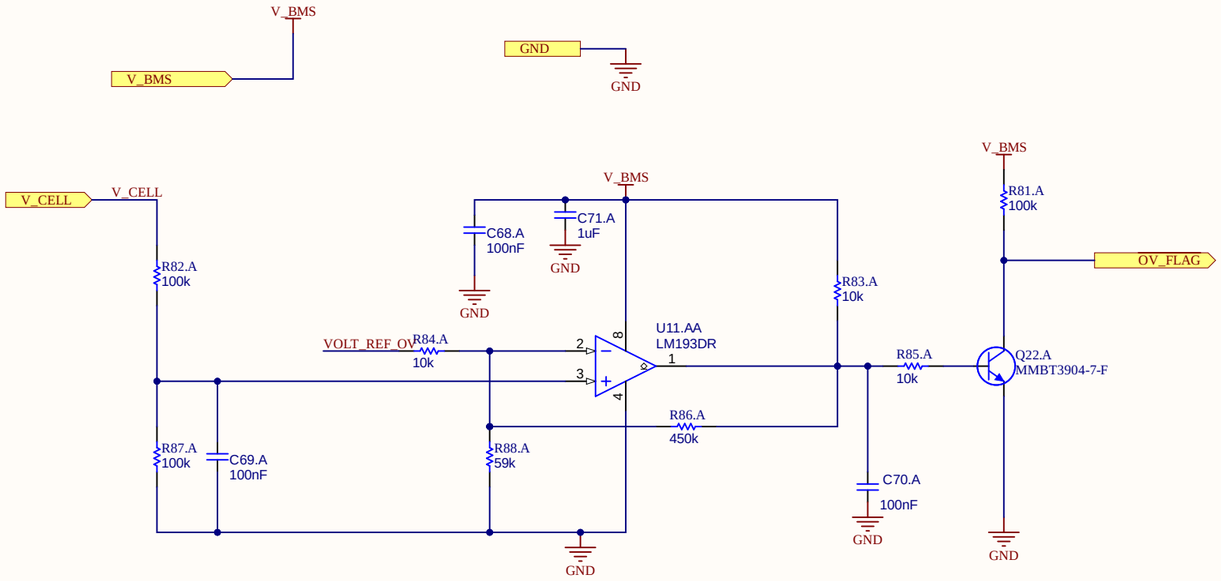
Will use 2 ideal diode controllers

SPACE CONCORDIA

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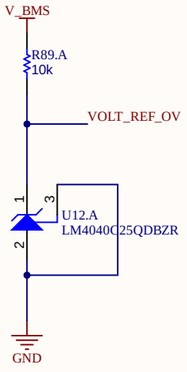
CIRCUIT
BMS BOARD

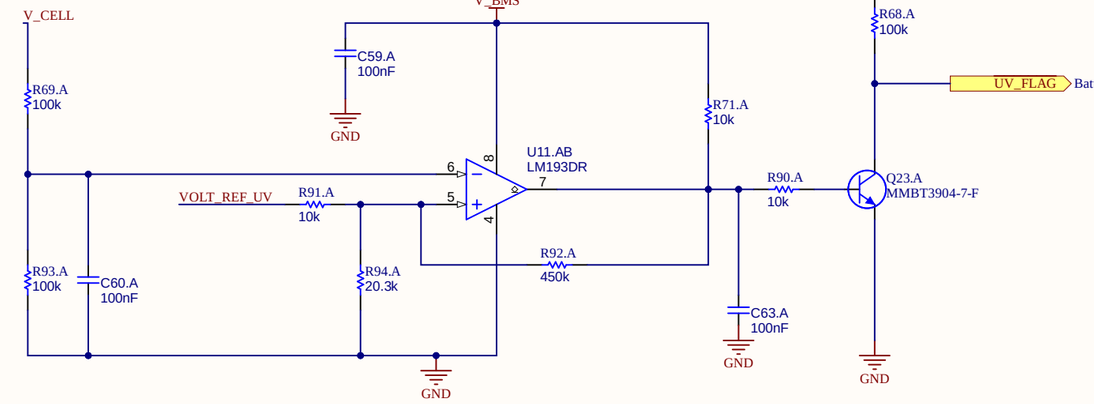
SIZE B | REV A | SHEET 14 / 19

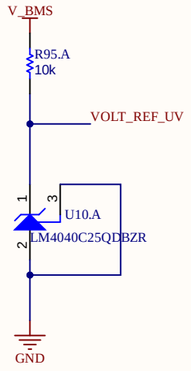


NOT YET CONFIRMED
VALUES:
VoltRef = 2.5V
OVPmax = 4.2V
Vbatt = UVP*2

V hyst = 4.05V







VALUES:
VoltRef = 2.5V
UVPmin = 3.3V
Vbatt = UVP*2

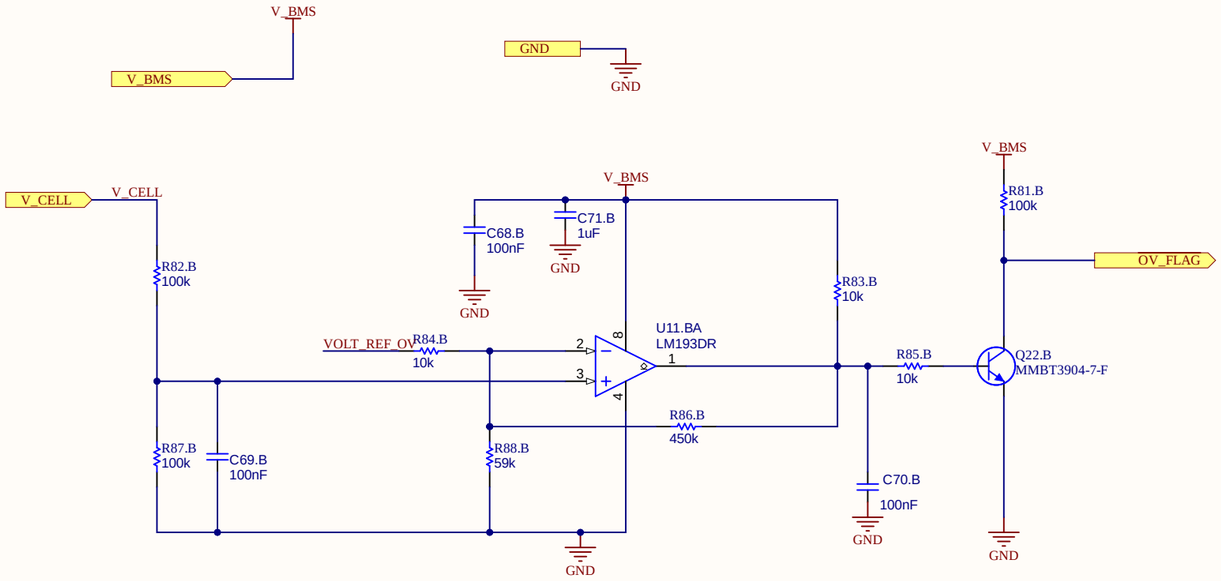
V hyst = 3.44V

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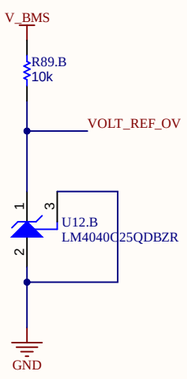
CIRCUIT
BMS BOARD

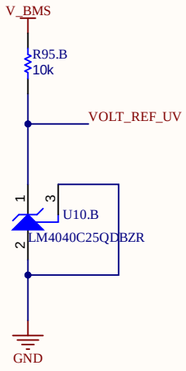
SIZE B | REV A | SHEET 15 / 20

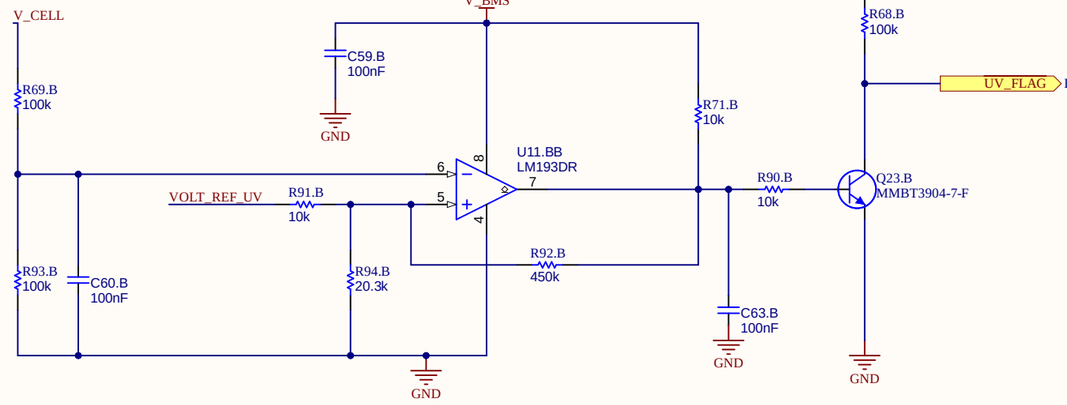


NOT YET CONFIRMED
VALUES:
VoltRef = 2.5V
OVPmax = 4.2V
Vbatt = UVP*2

V hyst = 4.05V



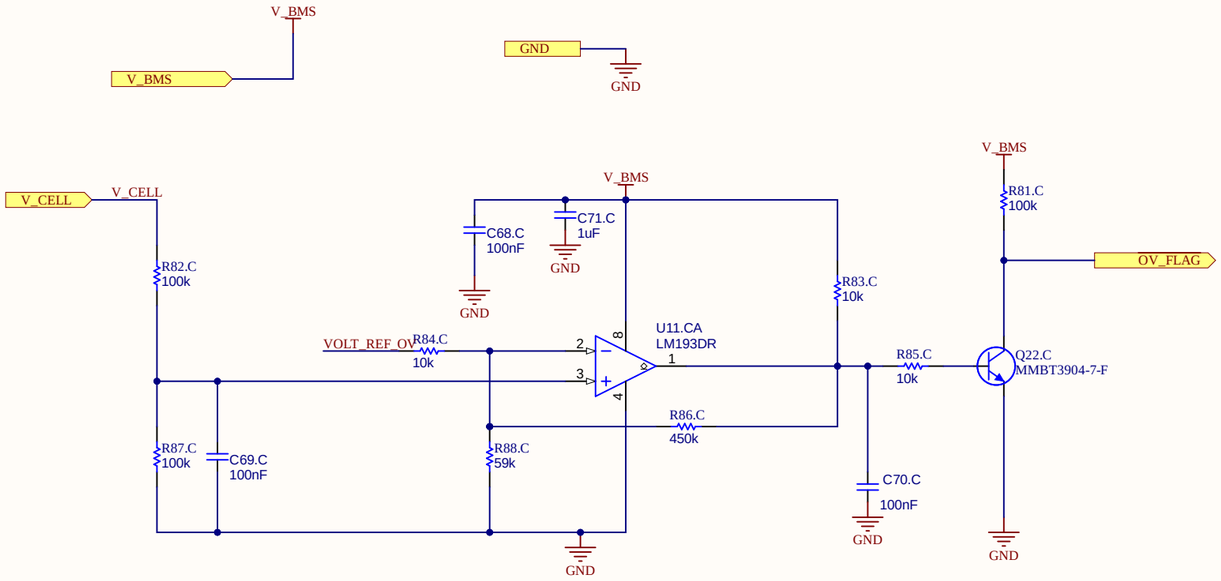




VALUES:
VoltRef = 2.5V
UVPmin = 3.3V
Vbatt = UVP*2

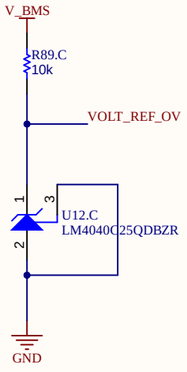
V hyst = 3.44V

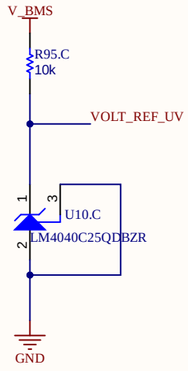
SPACE CONCORDIA			
TITLE *			
CIRCUIT BMS BOARD			
SIZE B		REV A	SHEET 15 / 20

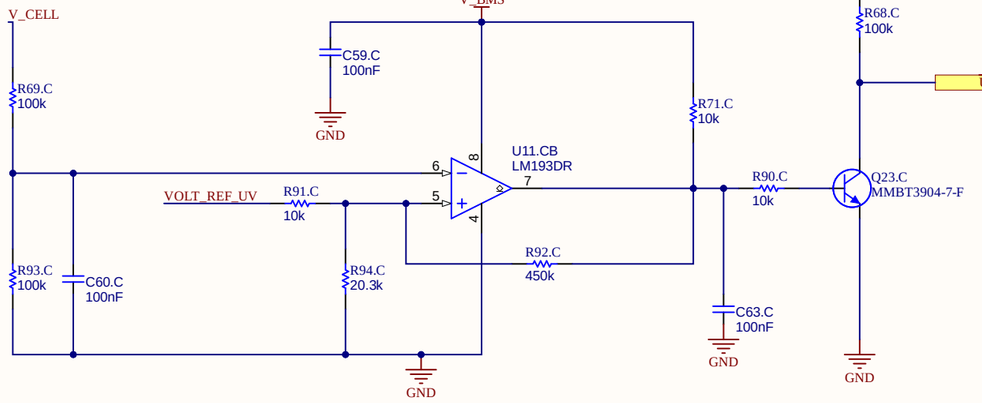


NOT YET CONFIRMED
VALUES:
VoltRef = 2.5V
OVPmax = 4.2V
Vbatt = UVP*2

V hyst = 4.05V



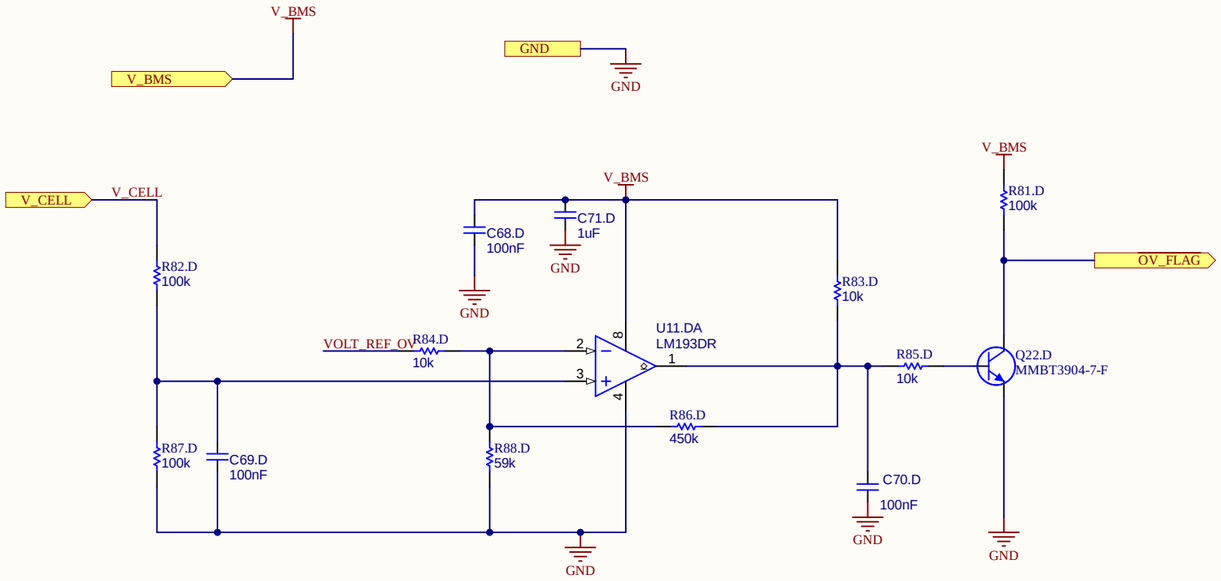


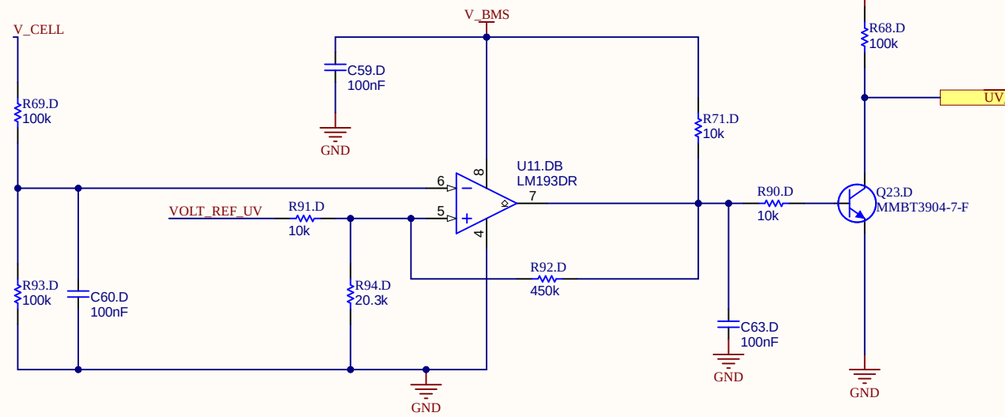


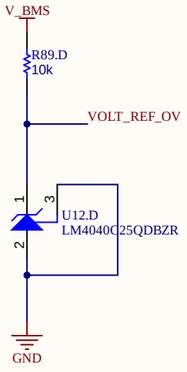
VALUES:
VoltRef = 2.5V
UVPmin = 3.3V
Vbatt = UVP*2

V hyst = 3.44V

SPACE CONCORDIA			
TITLE *			
CIRCUIT BMS BOARD			
SIZE B		REV A	SHEET 15 / 20

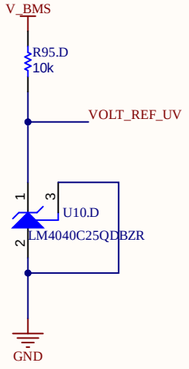






NOT YET CONFIRMED
VALUES:
VoltRef = 2.5V
OVPmax = 4.2V
Vbatt = UVP*2

V hyst = 4.05V



VALUES:
VoltRef = 2.5V
UVPmin = 3.3V
Vbatt = UVP*2

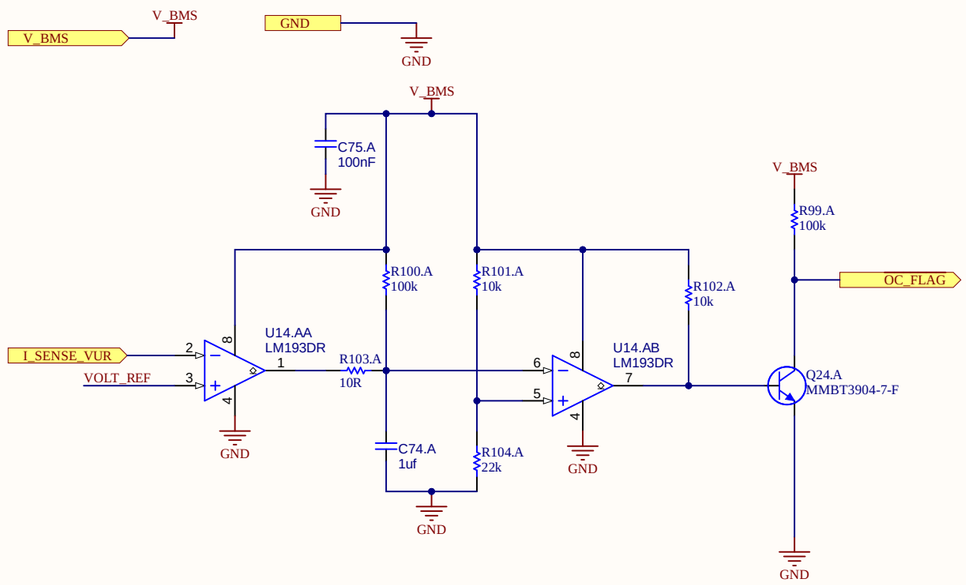
V hyst = 3.44V

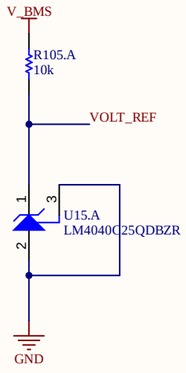
SPACE CONCORDIA

TITLE
*

CIRCUIT
BMS BOARD

SIZE B | REV A | SHEET 15 / 20



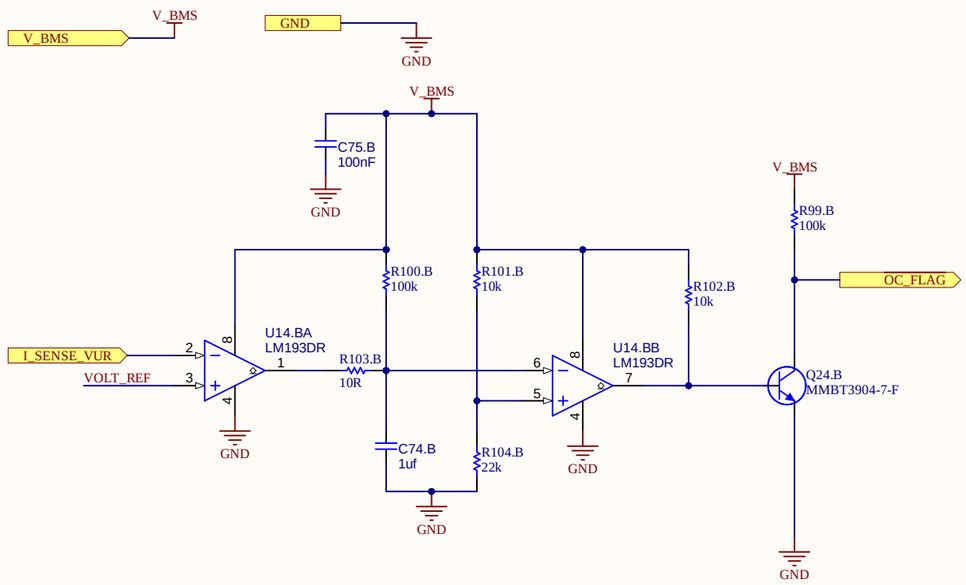


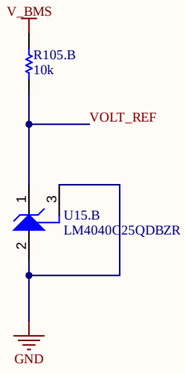
SPACE CONCORDIA

TITLE
*

CIRCUIT
BMS BOARD

SIZE B | REV A | SHEET 16 / 19



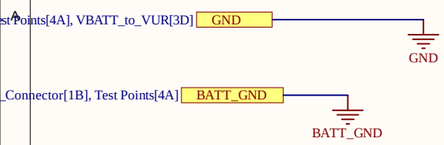


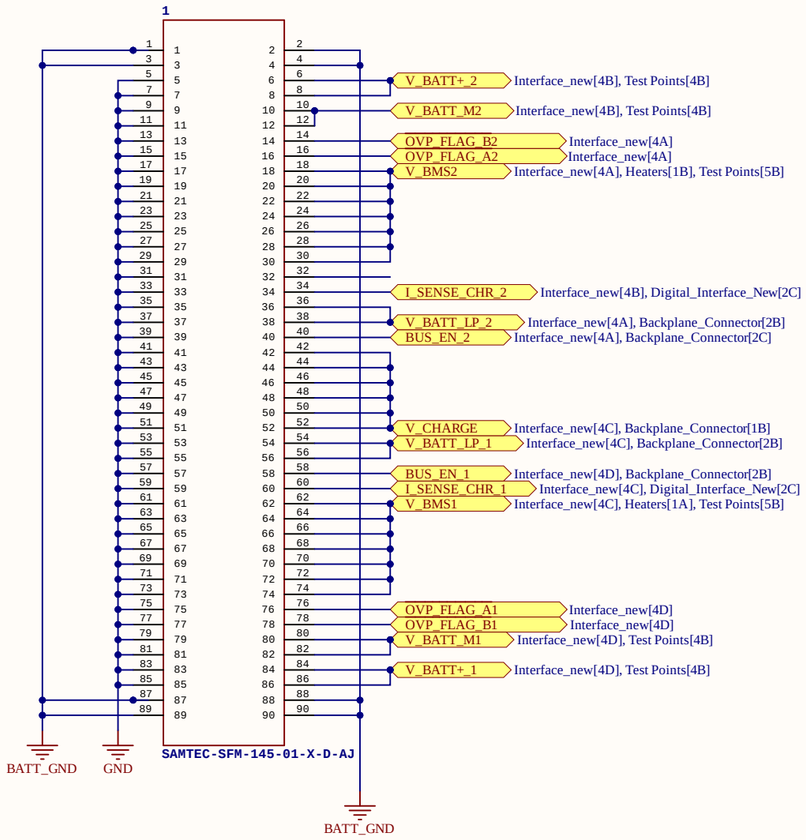
SPACE CONCORDIA

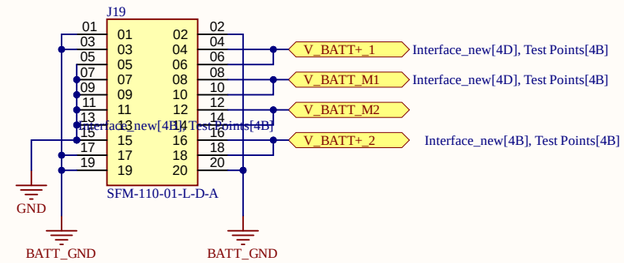
TITLE
*

CIRCUIT
BMS BOARD

SIZE B | REV A | SHEET 16 / 19





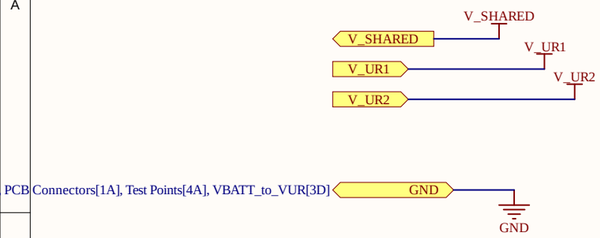


SPACE CONCORDIA

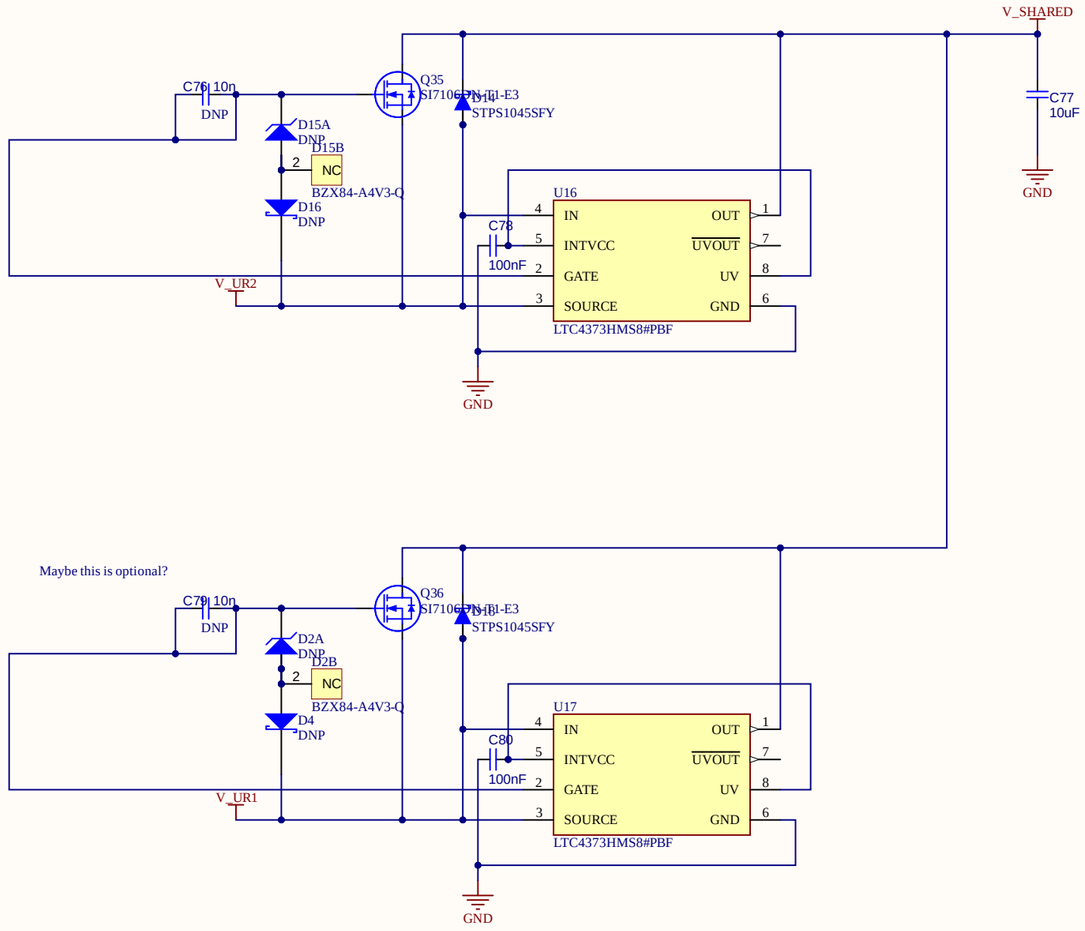
TITLE
*

CIRCUIT
BMS BOARD

SIZE B | REV A | SHEET 18 / 19

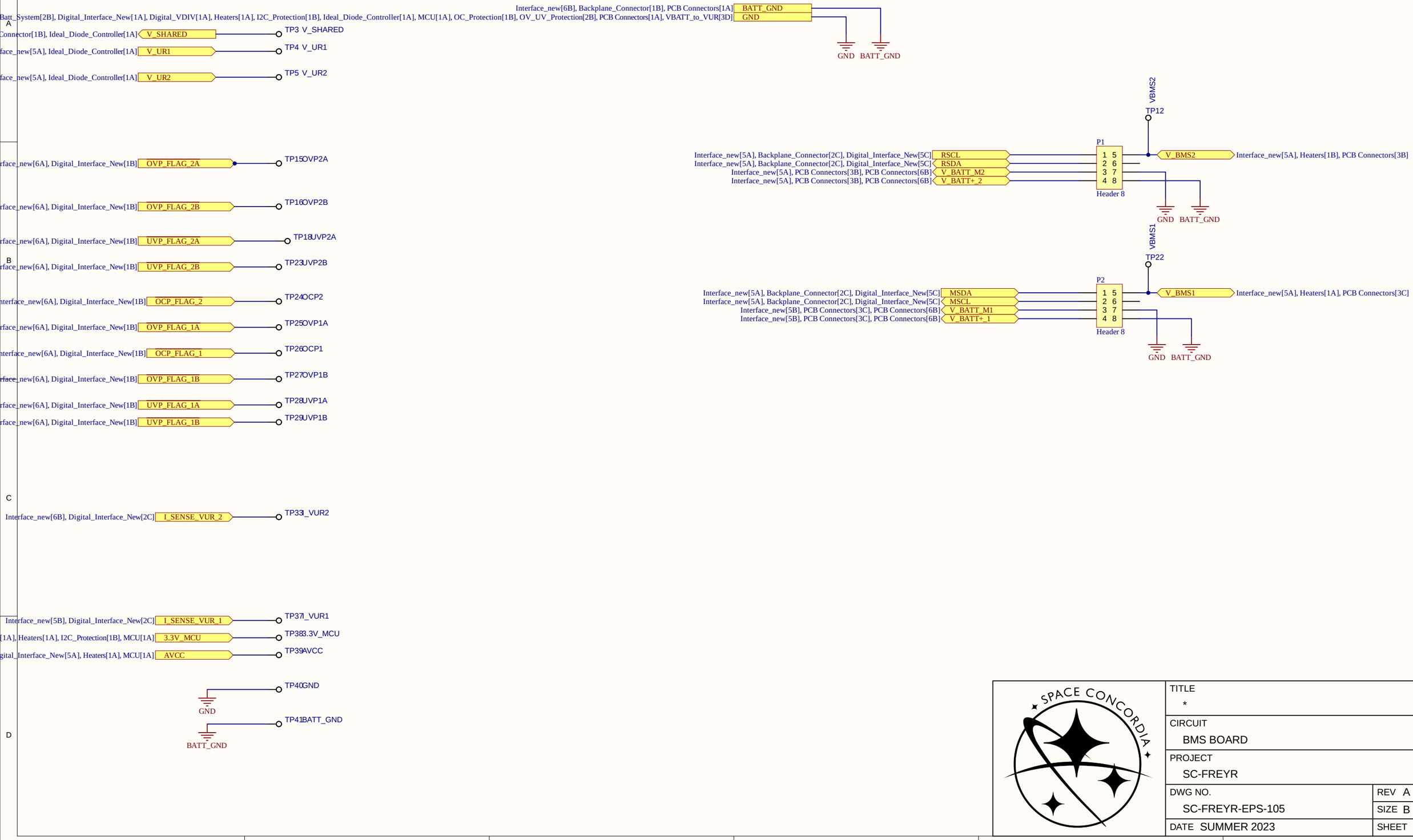


Just need D15 and R130, R129 can be removed, c76, r128 adn d13 can be dnp



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TITLE *			
CIRCUIT BMS BOARD			
SIZE B		REV A	SHEET 19 / 20

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TITLE	*
CIRCUIT	BMS BOARD
PROJECT	SC-FREYR
DWG NO.	SC-FREYR-EPS-105
REV	A
SIZE	B
DATE	SUMMER 2023
SHEET	21/21